

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	XXXXXXX
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed **Smart Lender** system uses machine learning to automate the loan approval process, making it faster, more accurate, and efficient. It reduces manual effort, improves decision-making, and enhances customer satisfaction. Key features include a **loan prediction model**, **real-time approval prediction**, and a **Flask-based web application**.

Project Overview	
Objective	Develop a machine learning-based Smart Lender system for quick and accurate loan approval prediction.
Scope	Build a Flask web application that predicts loan approval using applicant details and a trained ML model.
Problem Statement	
Description	Manual loan approval is slow, time-consuming, and may lead to inaccurate decisions.

Impact	Delayed approvals reduce efficiency and affect customer satisfaction
Proposed Solution	
Approach	Use a machine learning model with a Flask web application to automate loan approval prediction.
Key Features	• Real-time loan prediction • ML-based decision making • Simple and user-friendly interface



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 / Ryzen 5 (4 Cores)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	Flask

Libraries	Additional libraries	scikit-learn, pandas, numpy, pickle
Development Environment	IDE	VS Code / PyCharm
Data		
Model	Machine Learning Algorithm	Random Forest Classifier (Best Model)